Modelling Paracetamol Toxicity

Ioana Sarah Aizic

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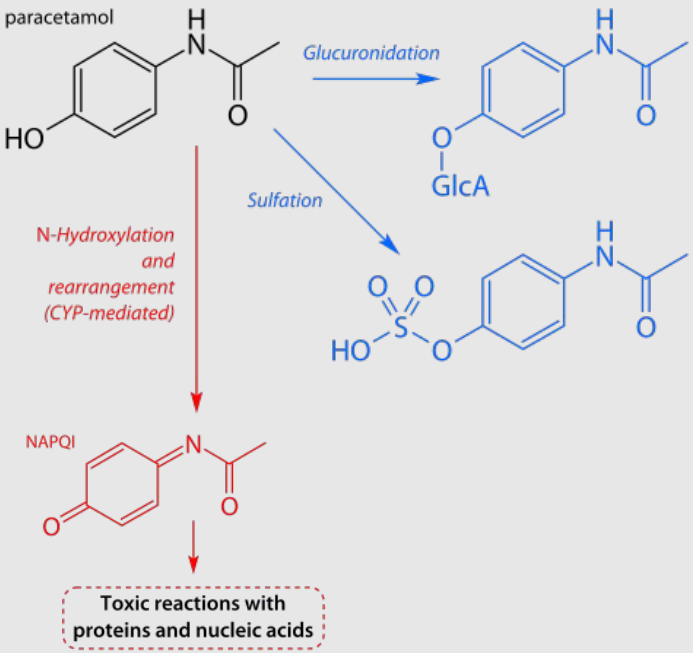
4. Limitations of the approach & future developments

Background



- Two non toxic pathway and one toxic pathway

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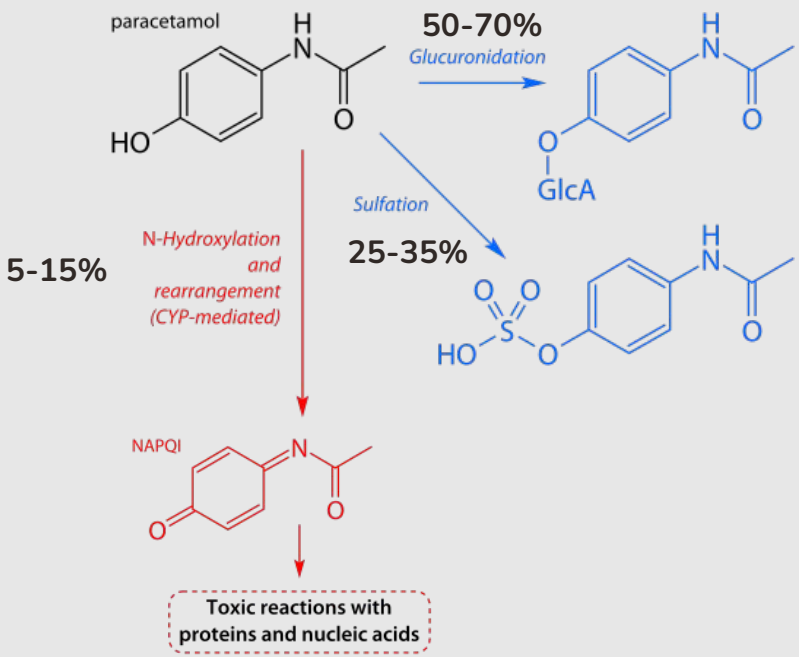

- Two non toxic pathway and one toxic pathway
- There is always a proportion of paracetamol which takes the toxic route

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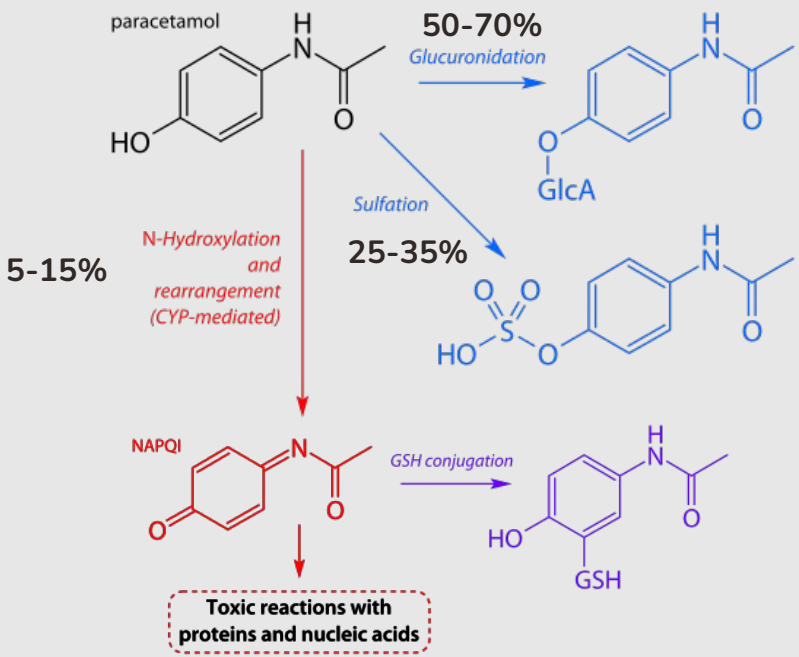

- Two non toxic pathway and one toxic pathway
- There is always a proportion of paracetamol which takes the toxic route
- Neutralised by GSH
- During overdose, GSH is depleted which is when NAPQI becomes toxic

Debated aspects of APAP toxicity

- How NAPQI leads to hepatocyte death
  - Downstream affected pathways are unknown

- Quantitative dynamics of APAP metabolism pathways
  - Competition between sulphation, glucuronidation, and oxidation in toxic / therapeutic ranges

- How toxicity affects other organs / pathways not liver-specific

Methods

1. Finding a transcripromic dataset
2. Preprocessing of the dataset
3. Making a tasklist
4. Adding missing metabolites, reactions, and gene rules to the model
5. Running the algorithm and mapping the results
6. Identify unexpected behaviours and pathways
7. Optimise model
8. Repeat steps 5, 6, and 7

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Article | Open access | Published: 04 February 2019

Identification of Novel Regulatory Genes in APAP Induced Hepatocyte Toxicity by a Genome-Wide CRISPR-Cas9 Screen

Katherine Shortt, Daniel P. Heruth, NiNi Zhang, Weibin Wu, Shipra Singh, Ding-You Li, Li Qin Zhang, Gerald J. Wyckoff, Lei S. Qi, Craig A. Friesen & Shui Qing Ye

Scientific Reports **9**, Article number: 1396 (2019) | Cite this article

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$$RPKM = \frac{Exon\text{-}Mapped\ Reads \times 10^9}{Total\ Mapped\ Reads \times Exon\ Length}$$

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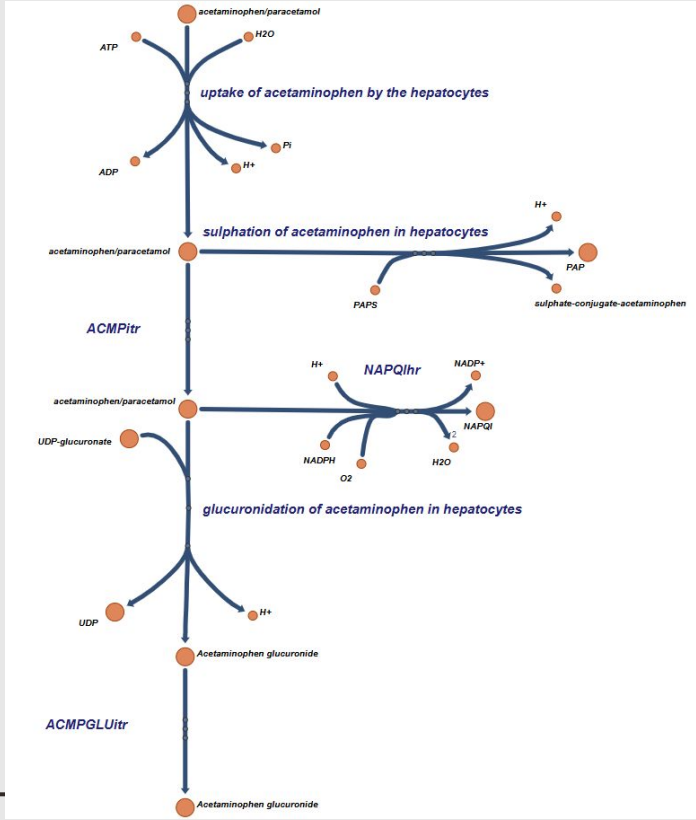
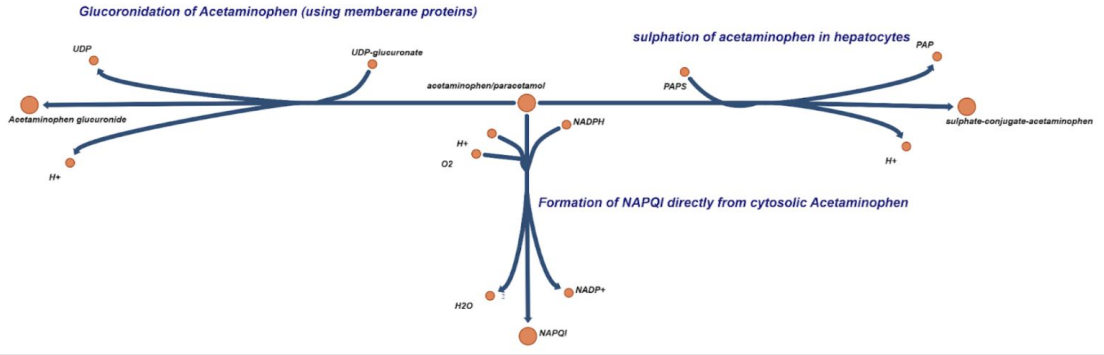
$$Min \sum_{i} \frac{yi,}{gi}$$

Methods: Model Modifications

Reactions	ID	Formula	Enzymes
Detoxification of NAPQI using GSH	MAR30000	*NAPQI + GSH → APAP-GSH*	GSTM1, GSTP1, GSTT1[1]
Formation of APAP-cys from APAP-GSH	MAR30001	*APAP-GSH + cysteine → APAP-cys + GSH*	GGT1, DPP4[2]
Formation of APAP-mer from APAP-GSH	MAR30002	*APAP-GSH + mercapturate → APAP-mer + GSH*	NAT1, NAT2[3]
Formation of APAP-mer from APAP-cys	MAR30003	*APAP-cys + mercapturate → APAP-mer + cysteine*	NAT1[3]

Methods: Model Modifications





Methods: Expression Dataset

Sample:	1	2	3	4	5	6	7
GeneID	T0	T30min	T3h	T6h	T12h	T24h	T4d
ENSG00000121410	60.10541125	53.69909983	57.70475052	60.06565419	67.28603652	65.00765254	96.4641602
ENSG00000148584	46.7450588	44.68613386	51.69164123	45.91349465	46.34359544	40.65421444	70.99830211
ENSG00000175899	41.08389252	50.89674907	42.40824444	46.56168517	45.46536402	46.59598426	31.12021386
ENSG00000166535	63.70267806	62.34851761	59.59307782	67.37940381	56.30138839	59.7851496	49.43522498
ENSG00000184389	28.30583145	30.78798883	27.53371094	26.92691423	28.33985172	29.80657549	15.02868252
ENSG00000128274	44.09239803	48.1322679	43.99064162	47.04782805	42.25644161	41.3187545	36.32817316
ENSG00000118017	37.7033675	39.7252156	33.38858051	37.62205769	31.41366163	38.42605077	32.43814643
ENSG00000094914	30.74822035	33.28738276	36.26326872	33.48984317	32.76478686	35.94379826	33.30968245
ENSG00000081760	43.13808714	42.45182715	35.97316256	41.37616106	47.52583001	44.07464116	32.31060457
ENSG00000114771	30.21445324	31.58325054	35.5248167	33.57086698	28.77896743	30.04111904	29.52594061
ENSG00000197953	61.70671258	59.98545426	60.89591816	66.25047199	66.27269259	63.20948536	51.03800101
ENSG00000188984	33.48175492	31.05307607	38.87422407	36.67677985	35.87237489	36.43243064	61.85780215
ENSG00000204518	41.84410626	39.46012838	38.71598435	41.97033571	43.74267935	37.56605779	37.05091036
ENSG00000109576	49.00952532	48.17013751	51.19054879	51.9092569	55.531247	52.92865998	76.61014396
ENSG00000103591	57.97034283	53.66123024	57.25640464	55.47430472	60.05751652	61.7826788	36.66828478
ENSG00000115977	31.78340503	27.72055083	29.51170742	29.19558102	35.02792161	30.7056591	38.00747431
ENSG00000087884	42.89546572	44.19382899	45.38842579	46.93979631	40.56753507	40.28285384	78.54452885

Methods: Tasklist

1. GSH de novo synthesis (minimal substrates, physiological excretion)	6. Paracetamol degradation via formation of APAP-mer (removed sulphation path) - flexible	11, Formation of PAP from cytosolic acetaminophen (sulfation) - fixed
2. Expanded GSH de novo synthesis (cyclic) - fixed	7. Formation of NAPQI from extracellular paracetamol (toxic route) - fixed	12. Formation of acetaminophen glucoronide from cytosolic acetaminophen (glucoronidation) - fixed
3. Paracetamol degradation via formation of APAP-cys - flexible	8. Formation of PAP from extracellular acetaminophen (sulfation) - fixed	13. Arachidonic acid metabolism(LOX pathway)
4. Paracetamol degradation via formation of APAP-mer - flexible	9. Formation of acetaminophen glucoronide (glucoronidation) - fixed	14. Arachidonic acid metabolism (CYP pathway - making 8hete)
5. Paracetamol degradation via formation of APAP-cys (removed sulphation path) - flexible	10. Formation of NAPQI from cytosolic acetaminophen (toxic route) - fixed	15. Arachidonic acid metabolism (CYP pathway - making 8hete)

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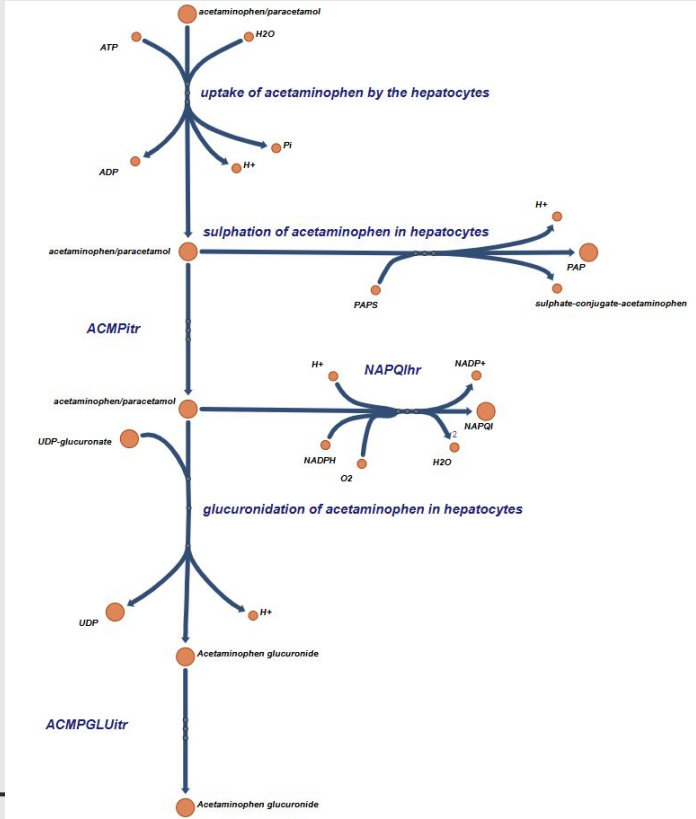
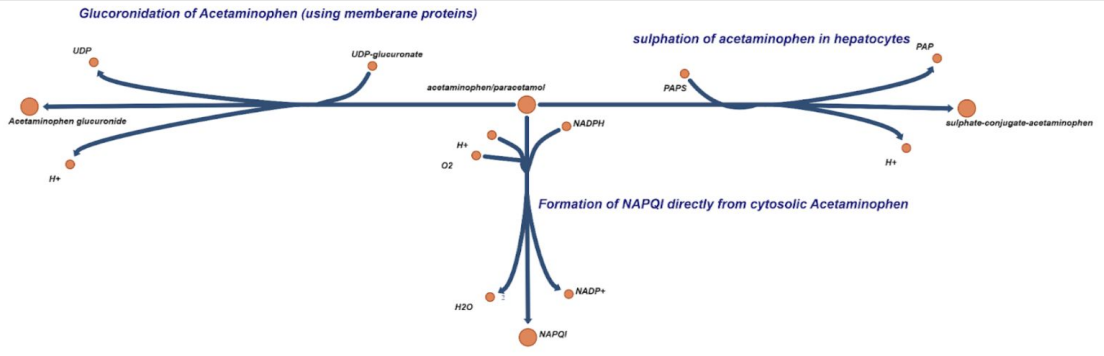
Results and Discussion



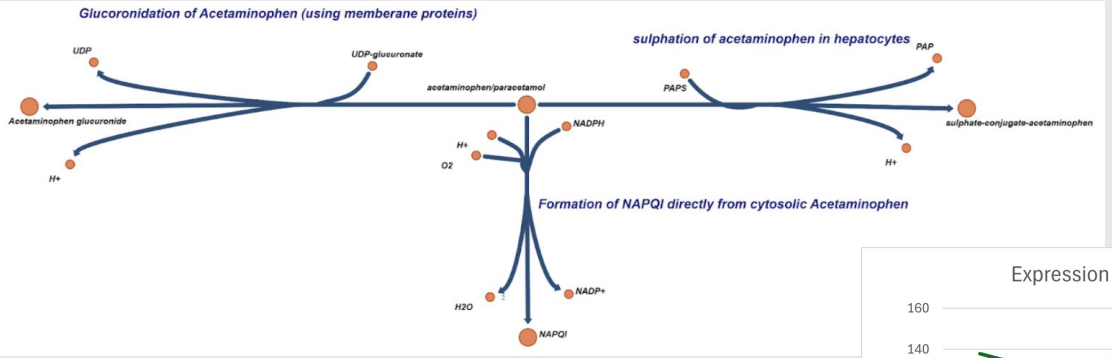
- This task can have 3 different endpoints
- The most “optimal” one is chosen based on the gene expression
- Strong preference for sulphation
- When sulphation was removed, more biologically correct choices were made

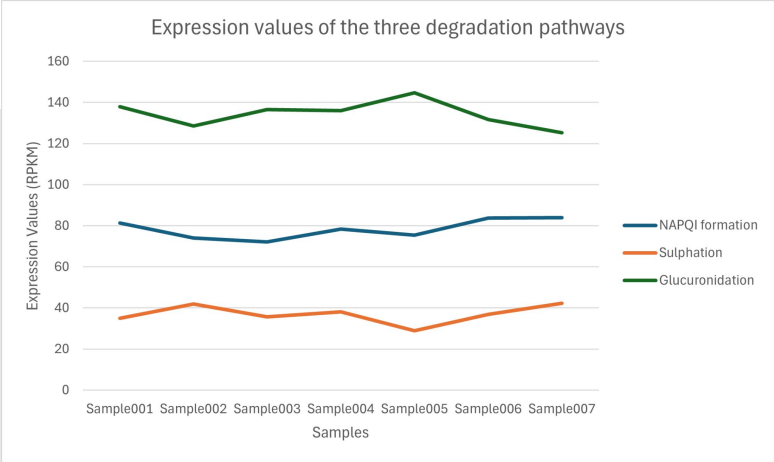
Results and Discussion



- Glucuronidation → glucuronosyltransferases
- Sulphation → sulfotransferases
- NAPQI formation → CYP450, specifically CYP2E1
- Only glucuronidation “preferred”

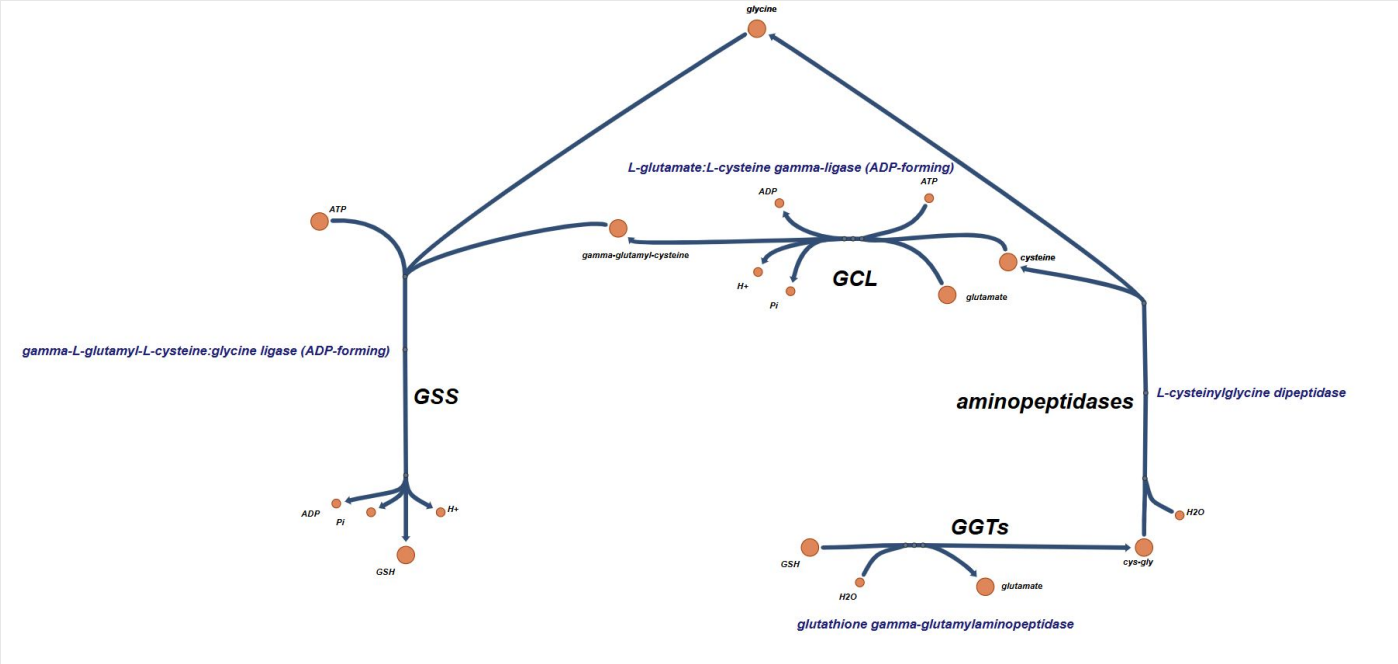
Results and Discussion





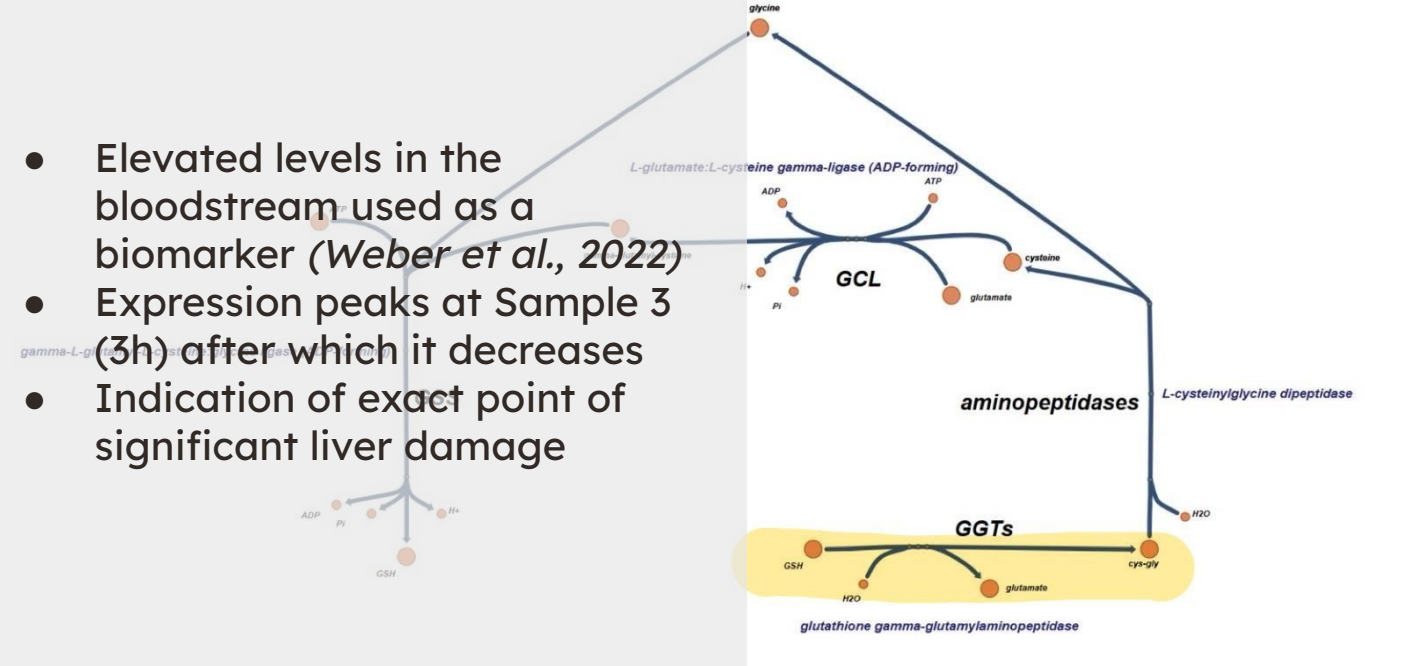
- Beginning of treatment with APAP → sulphation
- Sample 5 → glucuronidation
- NAPQI pathway has steady expression

Results and Discussion



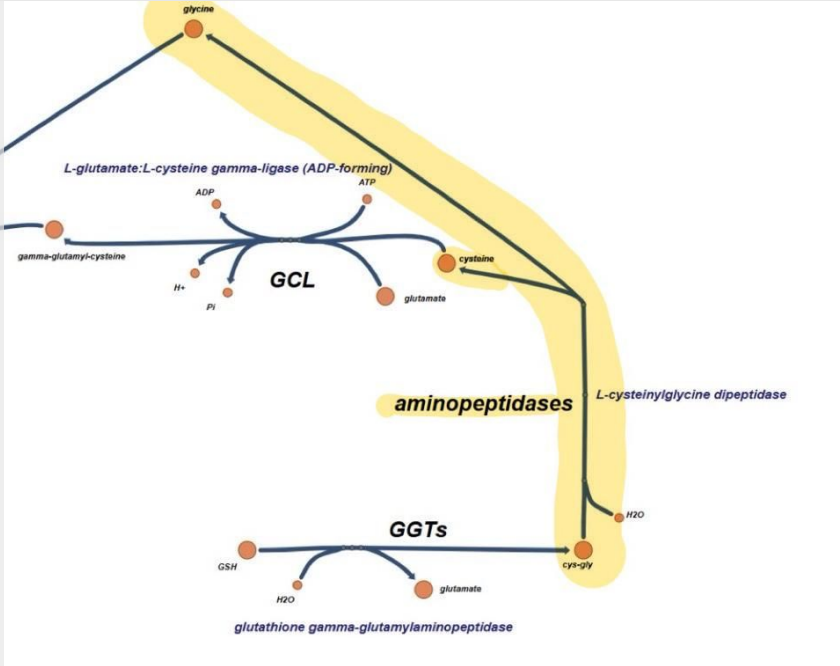
Results and Discussion

- Elevated levels in the bloodstream used as a biomarker *(Weber et al., 2022)*
- Expression peaks at Sample 3 (3h) after which it decreases
- Indication of exact point of significant liver damage



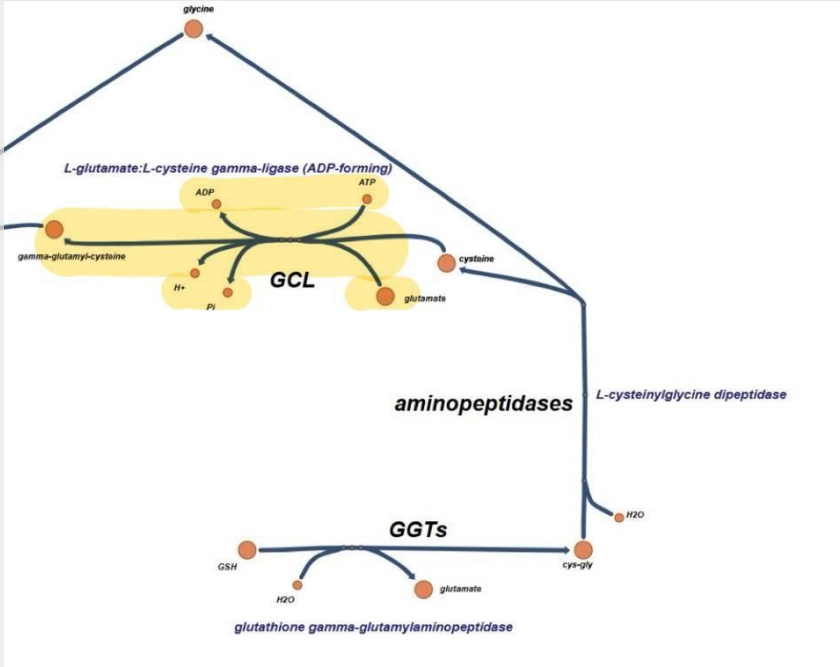
Results and Discussion

- Elevated levels in serum indicate liver damage *(Rutenburg et al., 1964)*
- Non-linear increase, increasing until 3h, decreasing after that, increase after 12h
- Expression peaks in Sample 7 (4d)



Results and Discussion

- Constant gene expression, until drop in Sample 7
- Cells with overexpressed GCL are more resistant to apoptosis *(Krejsa et al., 2010)*



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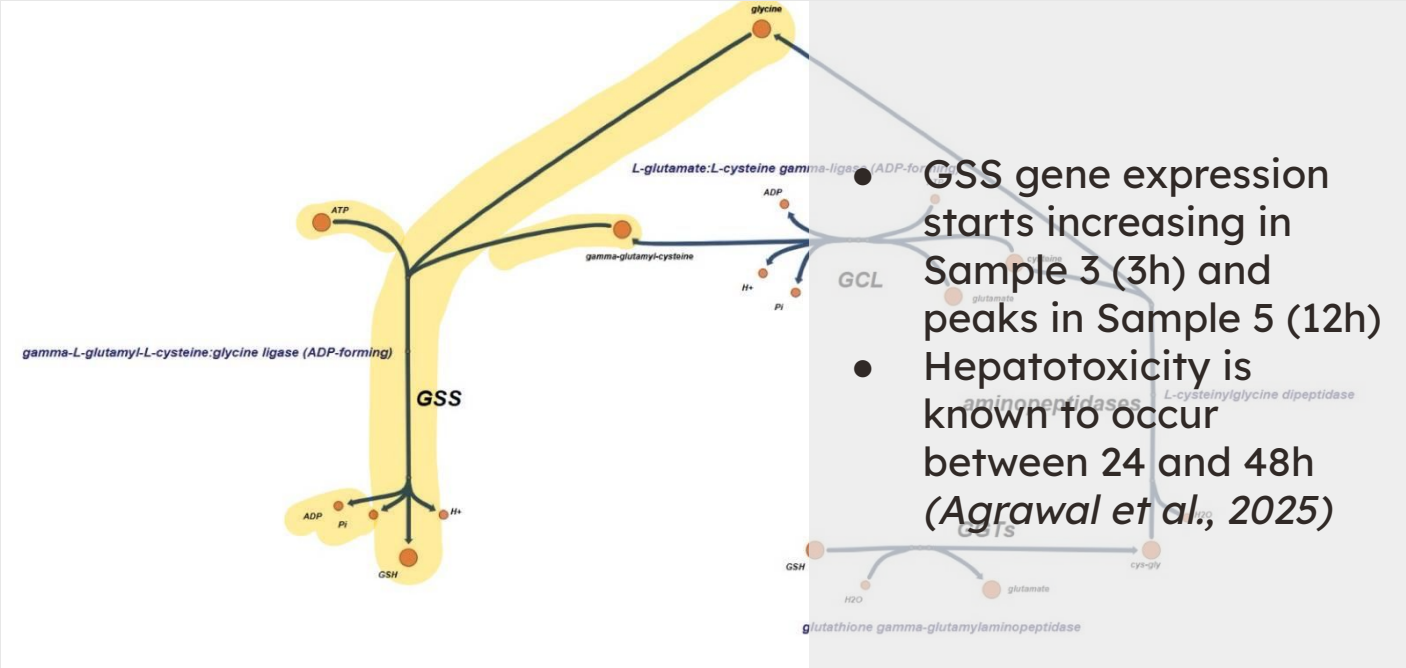
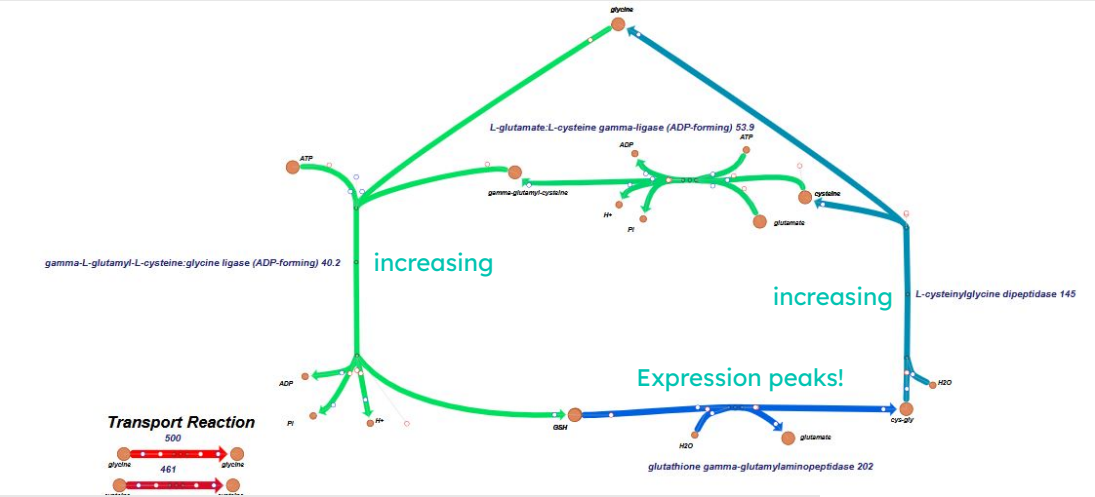


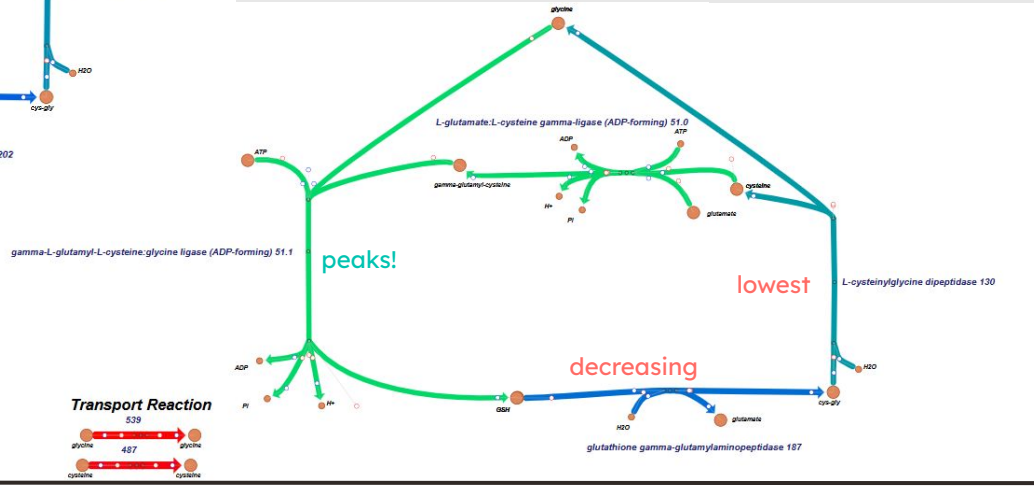
- GSS gene expression starts increasing in Sample 3 (3h) and peaks in Sample 5 (12h)
- Hepatotoxicity is known to occur between 24 and 48h *(Agrawal et al., 2025)*

Results and Discussion

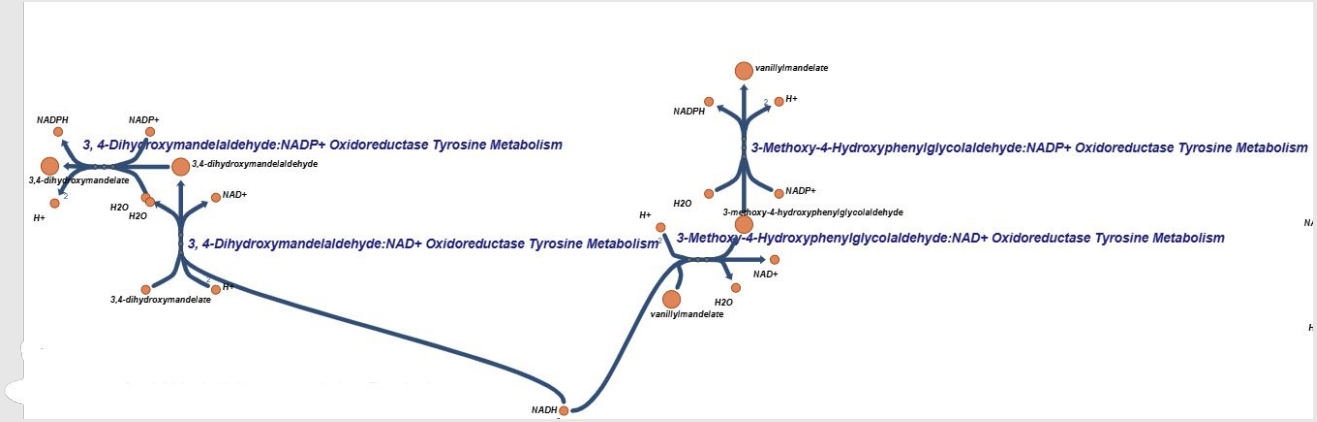


Sample 3 (3h)

Sample 5 (12h)

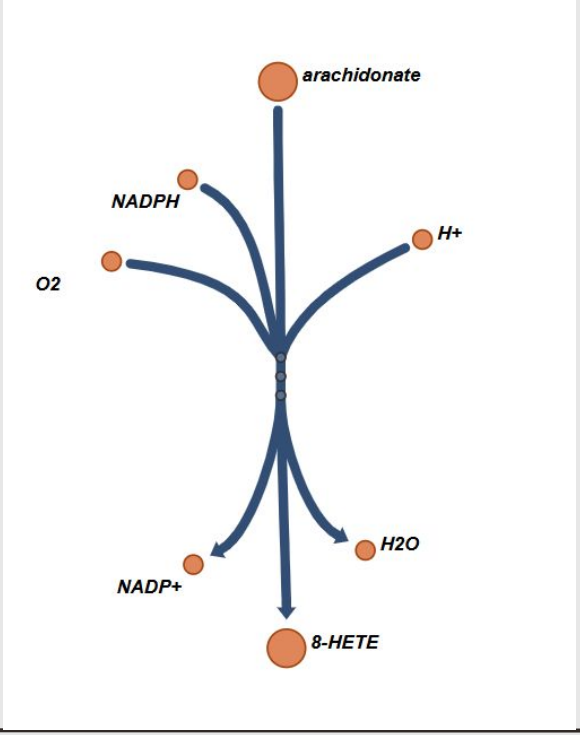


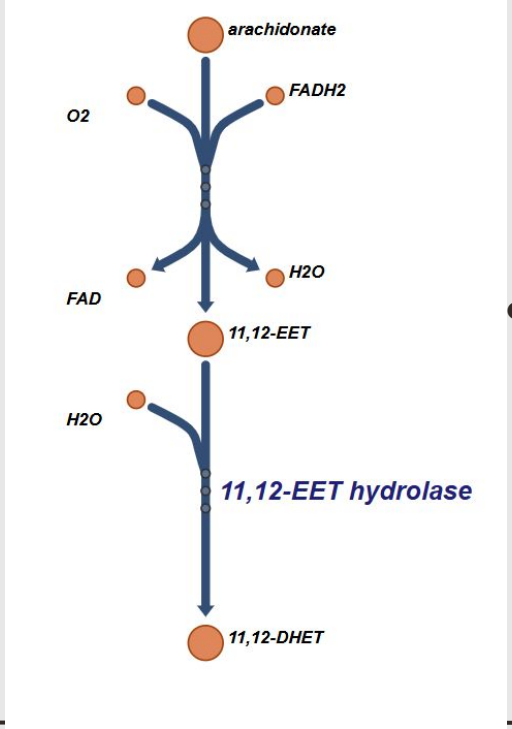
Results and Discussion: TCA cycle



- Healthy sample: dihydroxymandelaldehyde; paracetamol treated samples: vanillylmandelic acid
- ALDH enzymes upregulated with paracetamol exposure *(Wimborne et al., 2020)*

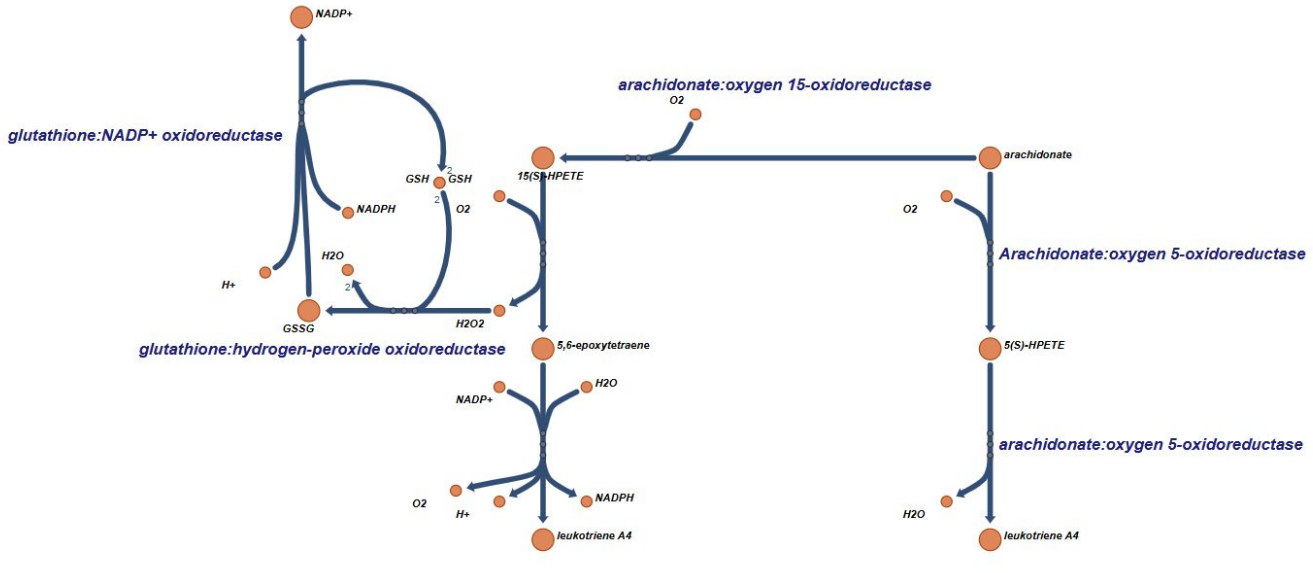
Results and Discussion: AA metabolism





- Possible fates of arachidonate in the CYP pathway
  - Monohydroxyeicosatetraenoic acid (HETE)
  - Dihydroxyeicosatrienoic acid (DHETE)
- Expression of CYPs highest in Sample 5 (peak toxicity) and second highest in healthy sample
  - Some isoforms inhibited by paracetamol *(Trettin et al., 2014)*

Results and Discussion



- Samples 1-6 choose the 5(S)-HPETE forming pathway *(Martínez-Clemente et al.)*
- Sample 7 chooses the anti-inflammatory 15(S)-HPETE producing pathway *(Wang et al., 2021)*
- Lipoxin A4 formation?

Limitations of such modelling & Future Developments

- Gene expression is an indirect measure of flux
  - Implement actual flux measurements
  - Test results of modelling in vivo / in vitro to verify
    - Use multiomics approaches
- No metabolite constraints
  - Implement metabolite levels and flux measurements into the model/algorithm
- Pathways chosen do not depend on biological accuracy

Thank you!!

Any questions?

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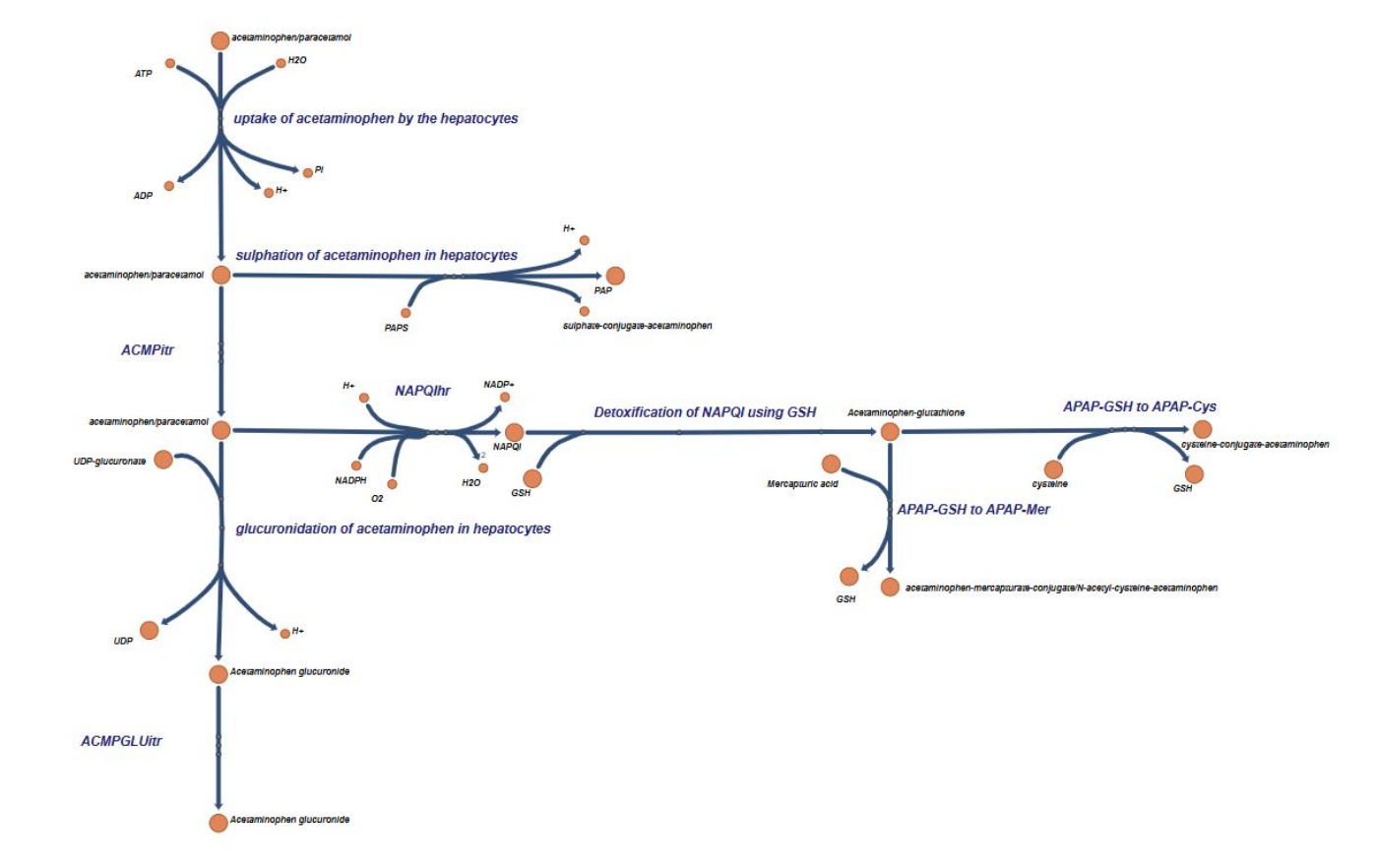
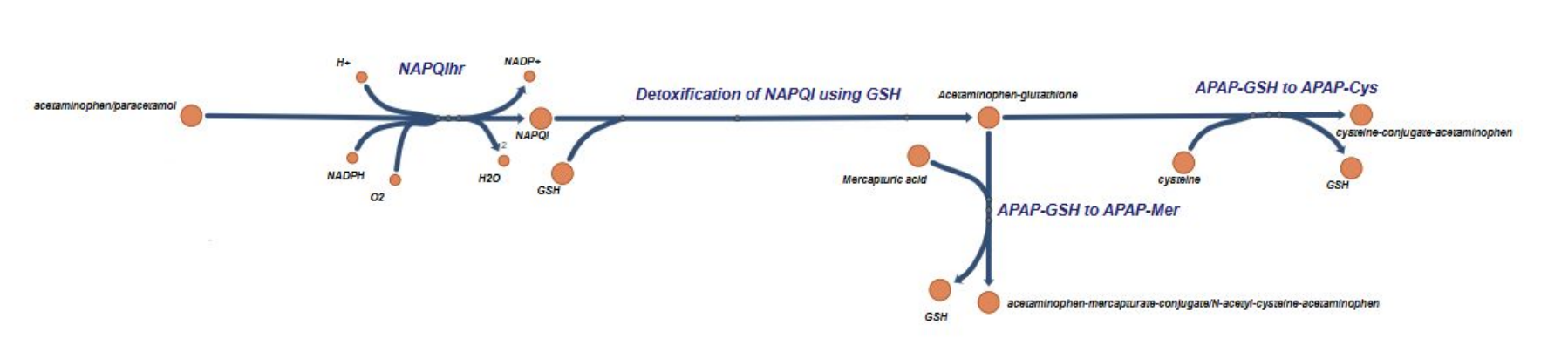

acetaminophen/paracetamol
ATP
H2O
uptake of acetaminophen by the hepatocytes
ADP
Pi
H+
sulphation of acetaminophen in hepatocytes
H+
acetaminophen/paracetamol
PAP
PAPS
sulphate-conjugate-acetaminophen
ACMPitr
H+
NAPQIhr
NADP+
acetaminophen/paracetamol
Detoxification of NAPQI using GSH
Acetaminophen-glutathione
APAP-GSH to APAP-Cys
NAPQI
cysteine-conjugate-acetaminophen
UDP-glucuronate
NADPH
O2
H2O
GSH
Mercapturic acid
cysteine
GSH
APAP-GSH to APAP-Mer
glucuronidation of acetaminophen in hepatocytes
GSH
acetaminophen-mercapturate-conjugate/N-acetyl-cysteine-acetaminophen
UDP
H+
Acetaminophen glucuronide
ACMPGLUitr
Acetaminophen glucuronide


acetaminophen/paracetamol
H+
NAPQIhr
NADP+
NADPH
O2
H2O
NAPQI
GSH
Detoxification of NAPQI using GSH
Acetaminophen-glutathione
Mercapturic acid
APAP-GSH to APAP-Mer
GSH
acetaminophen-mercapturate-conjugate/N-acetyl-cysteine-acetaminophen
APAP-GSH to APAP-Cys
cysteine
GSH
cysteine-conjugate-acetaminophen